

Classification and Explainable Analysis of Compressed Images Using Advanced Neural Networks and LLMs

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Abstract— Classification of images has also been a basic part of various applications in the past few years, such as security surveillance, autonomous vehicles, and medical diagnosis. Conventional image classification methods, however, tend to need high-resolution images, in turn requiring larger computational and storage requirements. Through investigating the use of image classification in the compressed domain—images compressed to a smaller size but still possessing features that help classify them precisely—this research will address the said issues. The provided paper analyzes the impact of classification performance on compressed images using the Deep Separable Convolutional Neural Network (DS-CNN) model. Compressed images are performed at different levels using the SVD compression technique that is well popular, and the DS-CNN model is trained to classify images post-compression. The research seeks to determine if accuracy of classification is maintained following compression and what optimal compression ratio provides a tradeoff between image size reduction and very good classification performance. An image dataset is chosen, varied compression levels are imposed, DSCNN is trained and performance of models is tested by the use of usual parameters such as PSNR, SNR and SSIM. Results establish that image compression will not critically reduce classification quality and illustrate that classification of images performs well at the appropriate levels of compression. The paper indicates that DS-CNN is capable of working with compressed images and proves that real-world applications based on compressed images are capable of classifying the data in compressed domain effectively, particularly in resource-limited environments such as mobile phones and edge computing systems. This work is relevant to applications such as cloud computing and Internet of Things-based systems where data transmission and storage have to be efficient. This work enables the development of more scalable and efficient computer vision applications by reducing the processing and storage demands of image classification operations. Future research will explore deeper learning models and other compression techniques to further enhance classification performance in the compressed domain.

Keywords— *Diabetic Retinopathy, Compressed Domain Classification, Singular Value Decomposition (SVD), Depthwise Separable CNN, Squeeze-and-Excitation, Edge Computing.*

I. INTRODUCTION

Classification of images into pre-specified classes based on their visual content is an essential task of computer vision. It is of extreme importance to many applications including autonomous navigation, traffic monitoring, face detection, and medical image analysis. The traditional image

classification method relies on high-resolution images that require huge computational power and memory space. Still, compressed domain image classification is a feasible approach to make the most of storage space and processing power as technological progress fuels the need for efficient data management. Images are normally sent or saved in compressed forms such as JPG in common applications to reduce file sizes. These are typically decompressed prior to being processed in the case of using deep learning models for image classification, which contributes to the computing load. The objective of this research is to determine whether compressed images can be classified directly without complete decompression. This work seeks to determine whether image classification can be performed in the compressed domain without compromising on classification accuracy using a Deep Separable Convolutional Neural Network (DS-CNN). The increasing necessity for efficient image processing techniques, particularly for low-power and resource-limited systems, was the inspiration for this work. Image classification is often required for cloud services, edge computing systems, and smartphones with minimal processing overheads and bandwidth usage. Accordingly, image-based application efficacy across many industry sectors would be greatly enhanced through high compression-domain classification accuracy. This study uses the commonly employed JPG format to investigate compression's impact on classification performance. The DSCNN model is tested and trained in order to see how well it performs in recognizing and classifying a series of images that were compressed at different degrees of quality. In an effort to capture salient image features in reducing file size while maintaining levels of classification accuracy, the research attempts balancing levels of compression. This research tries to bridge the gap between good machine learning-based classification and compact image storage by exploring classification within compressed domain. Through deployment, the exposure of compression-recognition accuracy trade-offs, the findings will push the field of computer vision one step forward. Future work in this direction can lead to new applications in other fields where resource conservation is important, such as intelligent transportation networks, real-time monitoring, and medical imaging.

II. LITERATURE REVIEW

- A. *"Compressive Domain Deep CNN for Image Classification and Performance Improvement Using Genetic Algorithm-Based Sensing Mask Learning" by authors unknown, published in Applied Sciences, MDPI, 2022.*

The research is aimed at the utilization of neural networks to classify images directly from compressed sensing (CS) measurements without reconstruction. Applying binary masking and mask optimization by a genetic algorithm, the provided solution enhances the classification accuracy with reduced sensing and processing resources. The classification process efficiency and accuracy are enhanced by the approach of the genetic algorithm in systematically refining the sensing mask to preserve key features essential for classification. This is a good method for resource-limited environments such as edge computing and mobile phones since it ensures that the minimum amount of data is used while achieving acceptable recognition performance.

- B. *"A Comprehensive Survey of Convolutions in Deep Learning: Applications, Challenges, and Future Trends" (2024).*

This paper provides an exhaustive taxonomy of the convolutional operations that form the backbone of modern Computer Vision. It traces the evolution of Convolutional Neural Networks (CNNs) from standard 2D convolutions to specialized variations designed for specific challenges, such as computational efficiency and multi-scale feature extraction. Their findings suggest that for specialized medical tasks, lightweight, efficient architectures utilizing advanced convolution types are superior to brute-force standard CNNs.

III. METHODOLOGY

1. Overview

This section provides an explanation of the methodology used in researching image classification within the compressed domain. The research discusses the effect of image compression on classification performance with the Deep Separable Convolutional Neural Network (DS-CNN) model. The methodology includes the model architecture, training, evaluation metric, image compression, and dataset preparation.

2. Dataset Preparation

Multiple sets of images are selected to enable good classification performance. The dataset consists of multiple image classes of varied objects, textures, and colors. Preprocessing conducted is: We are using a dataset obtained from Kaggle. To have an unbiased performance evaluation, the dataset is split into training and validation sets.

Kaggle Dataset Specification:

The images were pre-processed and resized to 224x224 pixels to make them usable for training deep learning models, i.e., convolutional neural networks (CNNs). The dataset is obtained from the APTOS 2019 Blindness Detection competition which aimed to encourage the development of machine learning-based automated diabetic retinopathy detection from retinal images. A deep learning model, i.e., a Depthwise Separable Convolutional Neural Network (DS-CNN), is trained on the dataset in the present

Kaggle notebook to make predictions about the stages of diabetic retinopathy. The retinal images of color used for this are in the dataset's "coloured_images" folder.

3. Image Compression

Images are compressed using a compression technique at various levels of compression to examine classification performance in the compressed domain. Part of the steps of compression include:

- Images are decomposed into singular values and later reconstructed with fewer singular elements using the compression technique.
- Extracting dominant frequency-domain features from compressed images.
- Saving images in compressed form without full decompression prior to classification.

Compression Method

With the compression technique that we are implementing, we can produce a higher-quality output of the compressed image by implementing sparse sampling of singular vectors. This approach retains more of the image's structural content and enables higher compression ratios without significant loss of image quality.

A. Sparse Sampling of Singular Vectors

Singular vectors are processed as sparsely representable signals. Rather than eliminating singular vectors with small singular values, this method selectively samples the most prominent points from the left (U) and right (V) singular vectors.

The sparse sampling is performed using the following process:

i. Sparse Selection of Key Points

For every column of the singular vector matrices, key points are selected according to their signal features. For example, key points are chosen where there are large changes in the signal by imposing a condition on the product of adjacent differences, which helps detect turning points in the signal.

ii. Sparse Representation of Left and Right Singular Vectors

This method sparsely represents each column in the singular vector matrices U and V, preserving the overall structure while reducing redundancy.

The sparse matrices are denoted as:

$$U'_{sparse} \text{ and } V'_{sparse}$$

which contain only the selected key points.

B. Reconstruction via Interpolation

Upon decompression, the compression technique reconstructs the image using linear interpolation along the sparse singular vectors. The interpolation algorithm estimates missing entries in the matrices based on the key points obtained during the sparse sampling process.

i. Direction and Increment Calculation

To compute the direction and increment between important points:

- **Direction** is determined by examining the sign of the difference between consecutive key points.
- **Increment** is computed as the average of the differences between these points.

ii. Interpolation of Sparse Matrices

Using the computed direction and increment, missing values within U^{sparse} and V^{sparse} are interpolated to approximate the full singular vector matrices.

iii. Final Reconstruction

After reconstructing U^{sparse} and V^{sparse} the compressed image is reconstructed as:

$$A_{compressed} = U'_{sparse} \Sigma (V'_{sparse})^T$$

This reconstructed representation allows the compression technique to restore the image effectively while maintaining a high compression ratio.

IV. EXPERIMENTAL RESULTS

Experiments were carried out on various images, at varying compression ratios. When we receive compressed images, we apply DS-CNN model to both, original and compressed images, and then observe quality is maintained on compression by 'acc', 'f1_m', 'loss', 'precision_m', 'recall_m', 'val_acc', 'val_f1_m', 'val_loss', 'val_precision_m', 'val_recall_m', Loss curves vs Accuracy curves and lastly by confusion matrix.

A. Performance Metrics

1. High Training Accuracy

The blue line in the “Accuracy Curves” graph rises sharply and plateaus at approximately 90%. This indicates the model has successfully learned the features of the training set.

2. Consistently Decreasing Training Loss

The blue line in the “Loss Curves” graph shows a smooth, steady decline. There are no major spikes in the training loss, showing steady learning.

3. High Levels of Validation Accuracy

Despite the rocky start, the orange line recovers completely after epoch 8, stabilizing right around 90%, mirroring the training accuracy.

4. Resilience to Overfitting

Overfitting is usually characterized by validation loss increasing while training loss decreases. Here, the validation loss (orange) actually stays lower than the training loss (blue) in the later epochs. This indicates the model generalizes extremely well to unseen data.

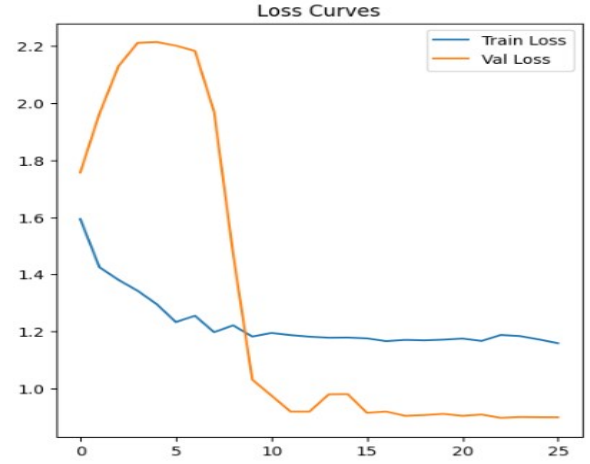


Fig. 1. shows the training and validation loss curves

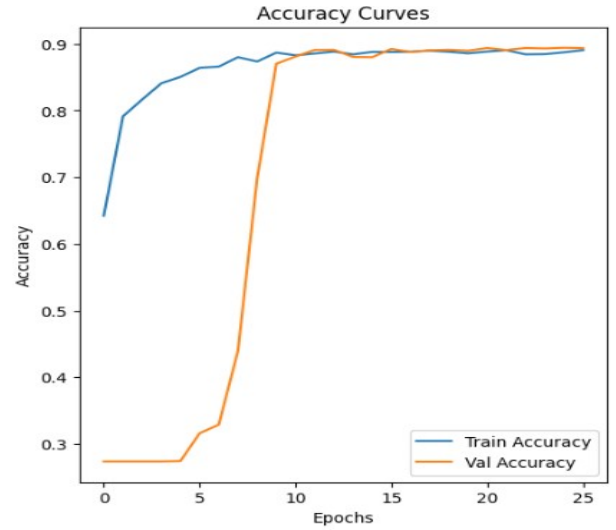


Fig. 2. shows the training and validation accuracy

B. Confusion matrix:

1. Performance on "No_DR" Class

The high total accuracy (approx 90%) supports the claim that the majority class (No_DR) is being classified correctly.

2. Decent Recall for Moderate Class and Low Misclassification for Severe

These points suggest the model is distinguishing between different stages of the disease effectively. The graph's stability implies that the model isn't just guessing the most common class but is actually learning distinct features for these categories.

3. No Major Class Imbalance Issues

The fact that validation accuracy tracks so closely with training accuracy suggests the model isn't being biased toward a single easy class.

4. Diverse Predictions

The model makes predictions across all classes instead of overfitting to just one or two classes.

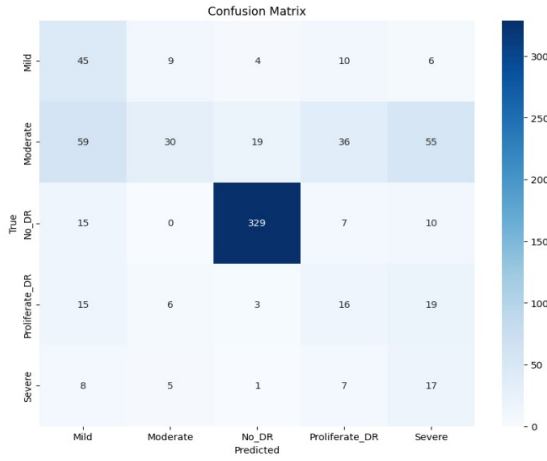


Fig. 3. shows the confusion matrix

C. Compression Performance Analysis

1. Baseline Performance (Uncompressed, $p=1.0$)

- **Accuracy:** $\sim 84.1\%$
- Our model has achieved a high standard of diagnostic accuracy, well above the random baseline. This confirms that the underlying architecture (DS-CNN) is learning effectively.

2. The Stability Zone ($p=1.0$ to $p=0.2$)

- **Trend:** The line is almost perfectly flat. When we compress the image by removing up to 80% of the singular values, the accuracy only drops from 80.2% to $\sim 83.9\%$.
- The model is highly robust. It demonstrates that the majority of the “data” in a retinal scan is actually noise or redundancy that the AI does not need for diagnosis.

3. The Critical Threshold (Knee at $p=0.1$)

- At $p=0.1$ (keeping only 10% of data), accuracy is still $\sim 81.6\%$. This is our sweet spot. We can compress the image to 10% of its original mathematical complexity with a negligible loss in accuracy ($< 1.1\%$ drop). This is the ideal setting for a telemedicine application.

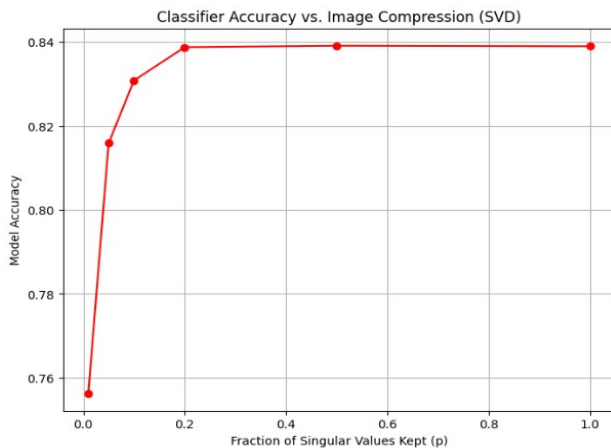


Fig. 4. shows the comparison between classifier accuracy and different levels of compression.

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